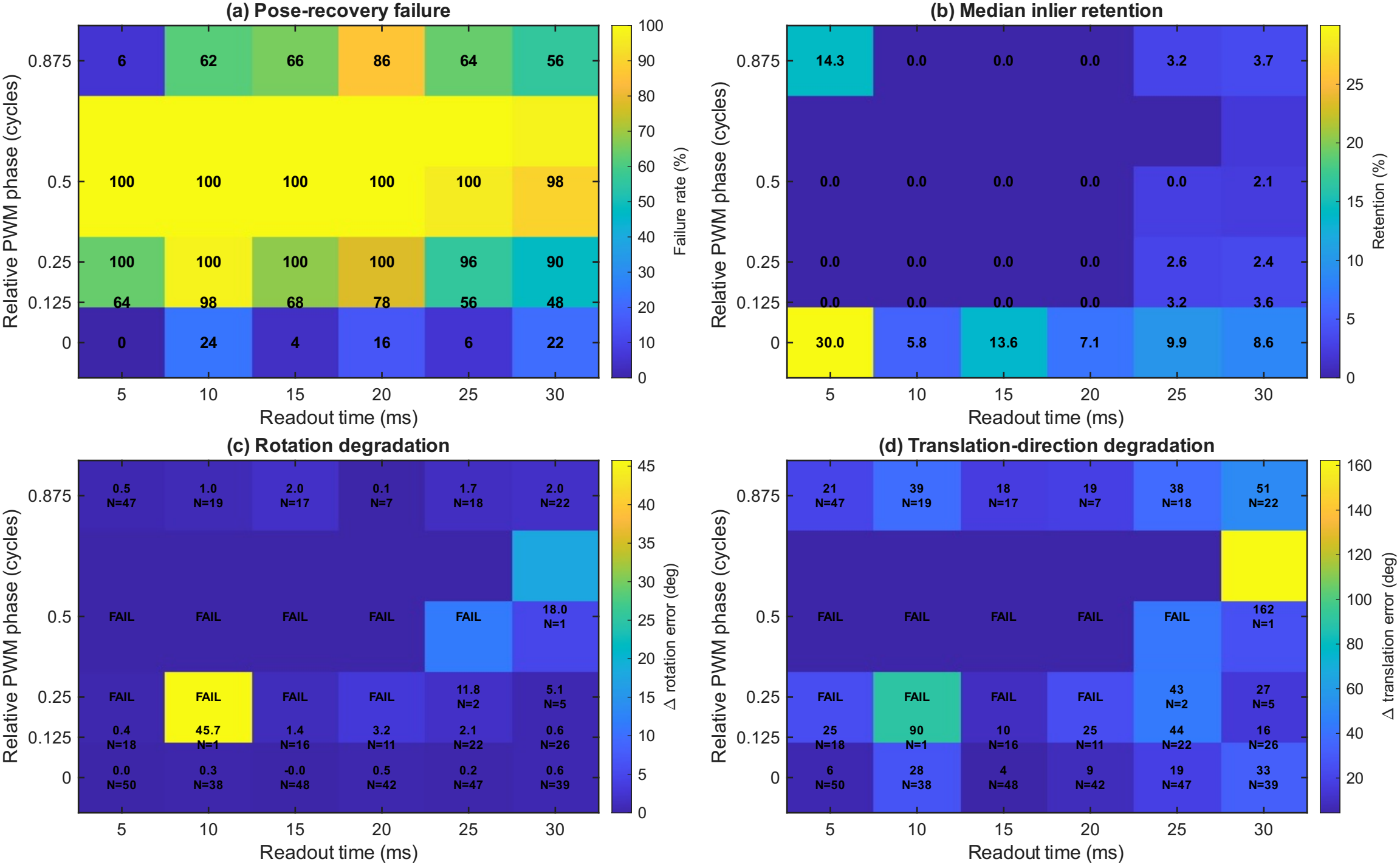

(a) Pose-recovery failure
Relative PWM phase (cycles)
0.875
0.5
0.25
0.125
0
6 62 66 86 64 56
100 100 100 100 100 98
100 100 100 100 96 90
64 98 68 78 56 48
0 24 4 16 6 22
5 10 15 20 25 30
Readout time (ms)
Failure rate (%)
100 90 80 70 60 50 40 30 20 10 0
(b) Median inlier retention
Relative PWM phase (cycles)
0.875
0.5
0.25
0.125
0
14.3 0.0 0.0 0.0 3.2 3.7
0.0 0.0 0.0 0.0 0.0 2.1
0.0 0.0 0.0 0.0 2.6 2.4
0.0 0.0 0.0 0.0 3.2 3.6
30.0 5.8 13.6 7.1 9.9 8.6
5 10 15 20 25 30
Readout time (ms)
Retention (%)
25 20 15 10 5 0
(c) Rotation degradation
Relative PWM phase (cycles)
0.875
0.5
0.25
0.125
0
0.5 N=47 | 1.0 N=19 | 2.0 N=17 | 0.1 N=7 | 1.7 N=18 | 2.0 N=22
FAIL | FAIL | FAIL | FAIL | FAIL | 18.0 N=1
FAIL | FAIL | FAIL | FAIL | 11.8 N=2 | 5.1 N=5
0.4 N=18 | 45.7 N=1 | 1.4 N=16 | 3.2 N=11 | 2.1 N=22 | 0.6 N=26
0.0 N=50 | 0.3 N=38 | -0.0 N=48 | 0.5 N=42 | 0.2 N=47 | 0.6 N=39
5 10 15 20 25 30
Readout time (ms)
Δ rotation error (deg)
45 40 35 30 25 20 15 10 5 0
(d) Translation-direction degradation
Relative PWM phase (cycles)
0.875
0.5
0.25
0.125
0
21 N=47 | 39 N=19 | 18 N=17 | 19 N=7 | 38 N=18 | 51 N=22
FAIL | FAIL | FAIL | FAIL | FAIL | 162 N=1
FAIL | FAIL | FAIL | FAIL | 43 N=2 | 27 N=5
25 N=18 | 90 N=1 | 10 N=16 | 25 N=11 | 44 N=22 | 16 N=26
6 N=50 | 28 N=38 | 4 N=48 | 9 N=42 | 19 N=47 | 33 N=39
5 10 15 20 25 30
Readout time (ms)
Δ translation error (deg)
160 140 120 100 80 60 40 20